



National Institute of Education Graduate Research Conference

2nd October 2026

Bridging Academia, Technology, and Industry in Education in the Age of AI

2nd October 2026, 08:30 – 17:00

National Institute of Education

Lecture Theatre 2



An Institute of



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Welcome Message



Dear Students and Colleagues,

I am excited to welcome you to the NIE Graduate Research Conference in October 2026.

This year, the theme of the conference is “Bridging Academia, Technology, and Industry in Education in the Age of AI”. The conference features high-quality graduate student research presentations. You can also watch exciting presentations of students participating in the Three Minute Thesis (3MT®) competition as well as interesting poster presentations and CrossSolve competition.

We greatly appreciate your participation in this graduate student event. As of 11 September 2026, we have registered more than 250 participants and presenters from Singapore, Hong Kong, South Korea, China, Indonesia, Chile, the UK, and the Philippines. In addition, students from Bangladesh and the USA will be joining us online.

The Office of Graduate Studies and Professional Learning prepared this event together with student leaders from diverse disciplines at NIE. We hope the conference will provide a platform for students to share and discuss their research, connect, and interact professionally.

I hope you have a rewarding conference experience!

Sincerely,
Associate Professor Cheung Yin Ling

Associate Dean, Research Student Development
Office of Graduate Studies and Professional Learning
National Institute of Education
Nanyang Technological University

Keynote Speaker



Associate Professor Caroline G. Lee

Vice Dean, National University of Singapore (NUS)
Graduate School

Co-Director, MSc in Precision Health and Medicine x AI

Associate Professor, Department of Biochemistry, NUS
Medicine, National University of Singapore

Associate Professor, Cancer and Stem Cell Biology
(CSCB), Duke-NUS Medical School

Biography

Associate Professor Caroline G. Lee is Vice Dean of the NUS Graduate School and Associate Professor in the Department of Biochemistry at the Yong Loo Lin School of Medicine, National University of Singapore. She is also Co-Director of the MSc in Precision Health and Medicine and an Associate Professor at Duke-NUS Medical School. A genomics and precision-medicine researcher, her work spans cancer genomics, pharmacogenomics, population genomics, and machine learning for disease and treatment-response prediction. She has extensive experience mentoring postgraduate researchers across biomedical science, medicine, engineering and computational disciplines, while building collaborations that connect fundamental discovery with clinical and industry applications. Through her graduate-education leadership, she champions transdisciplinary scholarship, responsible innovation, and the development of researchers who can cross boundaries with confidence. Her talk will draw on these experiences to explore how AI can transform the PhD from a thesis journey into a platform for meaningful real-world impact.

Keynote Address

Beyond the Shortcut: AI, Human Judgement and the Future of Graduate Study

How Master's and PhD Students Can Learn Deeper, Research Smarter and Create Knowledge That Matters

Abstract

Artificial intelligence is rapidly transforming how graduate students learn, search the literature, write, code, analyse data, generate hypotheses and communicate their work. It can compress weeks of effort into hours, widen access to expertise and make previously difficult questions more approachable. Yet speed is not understanding, fluent answers are not knowledge, and polished outputs are not scholarship.

Drawing on the Chinese Room thought experiment, this keynote examines what graduate education must preserve, redesign and strengthen in an AI-rich world. For Master's students, AI can support personalised learning, rapid upskilling and more sophisticated real-world problem-solving. For PhD students, it can accelerate discovery and expand the research frontier. But the central responsibilities of scholarship remain profoundly human: asking consequential questions, creating original and trustworthy knowledge, validating evidence, exercising ethical judgement and accepting responsibility for the impact of one's work.

The keynote argues that graduate education must move beyond teaching students merely how to operate AI tools. It must develop **AI-augmented scholars** who can think with AI without surrendering their thinking to it. As complex problems increasingly transcend disciplinary boundaries, students must also become **anchored integrators**—deeply grounded in one or two disciplines, sufficiently broad to understand other ways of knowing, and capable of synthesising them into new insights and novel solutions.

The challenge is not whether graduates will use AI. It is whether they will use it to take intellectual shortcuts—or to ask better questions, think more deeply and create knowledge that genuinely matters.

Conference Programme

Time	Item	Venue
08:00	Registration and Morning Tea	Outside LT1
09:00	Conference Opening by the Emcee: Kevin Gregory, committee member of GRC 2026	LT2
09:05	Opening Address by Professor Chang Chew Hung, Dean, Graduate Studies and Professional Learning, NIE	LT2
09:15	Keynote Address by Associate Professor Caroline Lee, Vice Dean, NUS Graduate School	LT2
09:55	Group Photograph Taking	LT2
10:00	3MT® Competition	LT2
11:30	Lunch, Poster Presentation, and Brew & Banter	Outside LT1
12:30	Oral Presentation Parallel Sessions	Tutorial Rooms and via Zoom
14:30	Afternoon Tea	Outside LT1
15:00	CrossSolve Showcase (Emcee: Jia Yichen)	LT2
16:00	Break and Lucky Draw	LT2
16:30	Award Presentation and Closing	LT2
17:00	End of Conference	LT2

Oral Presentation Parallel Session 1: AI, Pedagogy, and Learning (1A)

Session Number: 1A

Time: 12:30 to 14:30

Venue: TR701

1. Teaching Mass Communication Theory with Socrates Method Using Artificial Intelligence

Musfiah Saidah

Universitas Islam Negeri Syarif Hidayatullah Jakarta

Abstract: Mass communication theory is essential to communication education, yet foundational concepts such as media effects, normative theories, and audience reception are often experienced by undergraduate students as abstract, theoretical, and disconnected from their everyday digital media practices. Existing literature highlights the potential of dialogic pedagogy to strengthen learner-centered engagement and of artificial intelligence (AI) to provide personalized learning support. However, less attention has been given to how AI can be integrated with the Socratic Method to transform the teaching of communication theory from knowledge transmission into sustained critical inquiry. This teaching practice addresses this gap by examining how AI-mediated Socratic dialogue can support students in questioning, applying, and critically reflecting on mass communication theories. The study asks: (1) How can the Socratic Method and AI-mediated dialogue be integrated into the teaching of mass communication theory? and (2) How does this pedagogical approach engage students in critical inquiry and the application of theoretical concepts to contemporary media realities? The approach was implemented through an eight-week instructional sequence involving guided AI dialogue, theory application, role play, debate, peer review, reflective activities, creative media production, and dialogic assessment. Importantly, ChatGPT was positioned not as an answer machine but as a question generator that challenged students' assumptions and encouraged deeper inquiry, while instructor intervention maintained disciplinary rigor. Preliminary classroom reflections indicate that sustained questioning helped students perceive theories as living analytical frameworks rather than static doctrines. Students became more willing to test ideas, confront contradictions, articulate their own positions, and connect theories with phenomena such as TikTok virality, influencer branding, misinformation, AI in journalism, and political communication. The practice demonstrates that combining Socratic questioning with AI-assisted dialogue can offer a replicable learner-centered model for teaching abstract communication theories while strengthening critical reflexivity, theoretical application, and dialogic engagement.

2. AI-Enhanced Problem-Based Learning and Metacognitive Regulation in Science Education: An Integrated Bibliometric and Systematic Review

Rizka Novia Rohmawati

(Online Presenter)

Sekolah Indonesia Jeddah (Indonesian School in Jeddah)

Abstract: The integration of Artificial Intelligence (AI) into Problem-Based Learning (PBL) has generated growing interest in science education, yet the conceptual relationship between AI-enhanced instructional design, metacognitive regulation, and problem-solving outcomes remains fragmented across separate bodies of literature. Prior frameworks have assumed that AI functions effectively as a metacognitive co-regulator, but this assumption has rarely been tested against the broader empirical evidence base on PBL and AI in science learning specifically.

This study integrates bibliometric mapping and systematic review to examine how AI-enhanced PBL research has conceptualised metacognitive regulation in science education, and to identify the conceptual gaps between these two literatures. A structured search of Scopus and Web of Science, supplemented by Google Scholar, is being conducted following an adapted PRISMA protocol for screening and eligibility. Bibliometric mapping is performed using Biblioshiny to visualise thematic clusters and keyword co-occurrence patterns, while systematic qualitative synthesis clarifies conceptual mechanisms across the retrieved full-text studies.

Preliminary evidence from the literature reveals a striking asymmetry: recent meta-analyses report large and stable effects of PBL on students' critical thinking and problem-solving skills (effect sizes ranging from 0.80 to 1.40), whereas meta-analytic evidence on generative AI's impact on learning outcomes shows substantial effects on language proficiency, academic achievement, and motivation, but a negligible and statistically non-significant effect on metacognitive ability ($g^+ = 0.078$). Bibliometric evidence further indicates that AI and learning-analytics keywords have not yet become central themes within PBL/STEM research clusters, pointing to an underexplored intersection between these domains.

Based on these findings, this study proposes the AI-Scaffolded Metacognitive PBL (AS-MPBL) Model, repositioning AI not as an autonomous metacognitive co-regulator but as a conditional pedagogical amplifier whose effectiveness depends on structural integration within PBL's problem-solving syntax. The study offers theoretical and practical implications for designing AI-supported science learning interventions that deliberately scaffold metacognitive regulation rather than assuming it emerges automatically from AI use.

3. Beyond Correct Answers: Socratic Questioning and the Pedagogical Quality of AI-Generated Feedback in Mathematics Education

Hasim Turkoglu

(Online Presenter)

University of Strathclyde

Abstract: Generative AI tools offer the potential to provide students with immediate, personalised feedback. A significant proportion of the burgeoning literature on this topic focuses on academic writing, foreign-language learning, learning to code, or generating clinical scenarios for medical students, with little attention given to mathematics education. Moreover, the pedagogical quality of feedback generated by AI, its capacity to support students' thinking, and the extent to which it avoids providing direct answers have been examined only to a limited extent. I ask, "How can ChatGPT provide pedagogically meaningful, real-time educational feedback to maths students, and how does the quality of this feedback vary across different prompting conditions and student personas?" I claim that rather than the AI providing ready answers, the feedback generated should support students' mathematical thinking by asking Socratic questions, without immediately providing the solution or the next step in the process. Using ChatGPT-5, I conducted a pilot study under three conditions, focusing on the two student profiles I have encountered most frequently during my years as a teacher (students exhibiting conceptual misunderstandings or those seeking direct answers) and three types of maths problems (expanding algebraic expressions, percentage-ratio problems and adding fractions). Preliminary findings suggest that ChatGPT generally produced accurate and comprehensible feedback; however, just stating the student's level did not systematically enhance the pedagogical quality of the interactions with the AI. When instructed to use Socratic questioning, ChatGPT initially encouraged students to identify their own errors, but when faced with repeated demands for a direct answer, it eventually abandoned this approach and provided the solution. The results indicate that effective AI feedback depends not only on mathematical accuracy but also on careful prompt design that genuinely brings the student's thought process to light and enables them to develop their own solution and mathematical thinking.

4. The AI-Assisted READER Program and Its Implementation in Deep Learning to Enhance Junior High School Science Teachers' Competencies

Harianti

(Online Presenter)

Universitas Pendidikan Indonesia

Abstract: Indonesia's Deep Learning policy requires teachers to design meaningful, mindful, and joyful learning experiences, yet the 2022 PISA results show that only 34% of Indonesian students reach Level 2 scientific literacy, compared with a 76% OECD average, and the 2024 TALIS data indicate that teachers' use of artificial intelligence (AI) remains limited. At the same time, unguided use of generative AI risks fostering automation bias and cognitive offloading among students. This study aimed to develop, examine the implementation of, and evaluate changes in science teachers' AI competency through the AI-assisted READER Program (Read, Ask and Answer, Discuss, Evaluate, Reflect) integrated with lesson study. A Design and Development Research (DDR) approach was employed across three phases, namely needs analysis, design and development, and implementation, involving junior secondary science teachers from six subject-teacher forum (MGMP) branches in Subang Regency. The needs-analysis findings confirmed a gap between the AI-competency demands set out in the UNESCO AI Competency Framework for Teachers and actual classroom conditions, providing the basis for designing the five READER stages together with the accompanying training package and online learning system. The implementation of the training and the plan-do-see lesson study cycle across the six branches proceeded with a high quality of planning and enactment, as reflected in observation results and participant perceptions. Teachers' AI competency showed a statistically significant gain in the cognitive aspect from pretest to posttest ($t(45) = 6.51, p < .001, N\text{-Gain} = 0.51$, medium category), while all attitudinal dimensions aligned with the UNESCO framework likewise showed significant gains ($Z = 6.28, p < .001$) with a large effect size ($d = 1.65$). Among students, early indications emerged of strengthened critical-thinking and AI-application skills, although the ability to independently evaluate and critique AI output still requires further reinforcement. These findings indicate that the AI-assisted READER Program makes a genuine contribution to strengthening teachers' AI competency and holds promise for supporting the development of students' AI competency within the context of Deep Learning in junior secondary science education.

5. The Emotional Ecology of Peer Feedback in Higher Education EFL Writing: A Review

Rahmati Putri Yaniafari

English Language and Literature, National Institute of Education

Abstract:

To understand peer feedback more comprehensively, it is important to map what, why, and how emotions arise and change within its unique emotional ecology. This literature review aims to identify the types of emotional experience students' feel during peer feedback exchanges in EFL writing class within higher education context and examine how these emotions interact with contextual factors. From the analysis of 28 reviewed studies, it is revealed that students' mixed emotional experiences during peer feedback process—both positive and negative—are shaped by their internal emotional antecedents, social conditions, and pedagogical design. These reactions, together with cognitive processes, shape their revision behavior and eventually produce emotion-regulation outcomes. This will become the emotional output that influences emotional input of the next feedback round, shaping a unique emotional ecology in peer feedback process.

Oral Presentation Parallel Session 1B: AI, Pedagogy, and Learning (1B)

Session Number: 1B

Time: 12:30 to 14:30

Venue: TR702

1. A Narrative Literature Review of Higher Education Educators' Perceptions of AI Chatbots

Lim Mei Hui

Psychology and Child and Human Development, National Institute of Education

Abstract: Artificial intelligence (AI) chatbots have become increasingly prevalent across society and industry, including the higher education sector. Their integration into education is influencing teaching and learning by providing students with immediate access to information, personalised learning support, and interactive learning experiences, while also assisting educators with instructional tasks. Despite these potential benefits, the adoption of AI chatbots raises questions about their appropriate and responsible use in higher education. Consequently, AI chatbots have become a prominent topic of discussion among educators, students, policymakers, researchers, and the wider public. A growing body of research has examined higher education educators' perceptions of AI chatbots, highlighting their perceived applications, benefits, challenges, and ethical implications. However, limited attention has been given to synthesising similarities and differences in educators' perceptions across studies and educational contexts. Therefore, this narrative literature review aims to synthesise existing research on higher education educators' perceptions of AI chatbots by identifying recurring themes, areas of consensus and disagreement, and contextual factors that may contribute to variations in educators' perspectives. Relevant literature was identified through keyword searches, with subsequent searches informed by themes and issues that emerged during the reading and analysis of the literature. Selected articles were examined, and additional literature was considered where emerging findings warranted further exploration. By examining patterns across the literature, this review seeks to provide a more comprehensive understanding of how higher education educators perceive and engage with AI chatbots. The findings may inform institutional policies, AI implementation strategies, and professional development initiatives, while providing a basis for anticipating educators' perceptions in specific higher education settings.

2. When AI Writes Audit Reports: Rethinking Audit Education for the Digital Age

Jingya Li

Capital University of Economics and Business

Abstract: A preliminary content analysis of 15 entry-level audit job advertisements from Big Four firms (collected May – July 2026) reveals three patterns. First, while only 5 out of 15 postings explicitly mention “AI”, 9 reference digital tools or data analysis, and 7 specify technical skills such as Python, SQL, or Power BI, suggesting that AI-related competencies are often embedded in operational rather than branded language. Second, critical thinking and professional judgement appear in 12 of the 15 postings (80%), underscoring that the profession values human judgement alongside technical growth. Third, a review of audit-related course offerings at five Chinese universities shows that all five have introduced courses such as “IT Auditing”, “Big Data Analysis and Auditing”, and “Intelligent Accounting”. However, questions remain as to whether these courses are systematically integrated into core curricula, whether they teach hands-on tool use as opposed to conceptual awareness, and whether they explicitly cultivate the critical thinking needed to interrogate AI-generated outputs. These findings suggest that while audit education is responding to technological change, the depth and coherence of that response warrant closer scrutiny.

3. An Educational Tutorial on Conducting AI-Assisted Biomarker Reviews

Lee Jia Wei

National University of Singapore

Abstract: Conducting systematic reviews is important for synthesizing research evidence, yet the rapid growth of scientific literature has made the systematic review process increasingly resource-intensive and time-consuming. Artificial intelligence (AI) presents a promising approach to accelerating and supporting various stages of the systematic review workflow. Although there are existing AI tools that can assist with systematic reviews, there is a lack of educational tutorials that integrate multiple AI tools into a seamless, end-to-end workflow and demonstrate their effective and responsible use. Thus, our study aims to develop an integrated tutorial leveraging AI tools to assist with biomarker systematic reviews, and to evaluate its educational effectiveness.

Our AI-assisted biomarker review tutorial integrates ASReview and ChatGPT to support literature screening and data extraction. To assess its educational effectiveness, a pre-post educational study was conducted in a Precision Biomarkers course at the National University of Singapore. The study protocol was reviewed and approved by the National University of Singapore Departmental Ethics Review Committee. Students in the course participated in a lecture and hands-on activity introducing the biomarker review workflow. Pre- and post-class surveys were administered to assess the students' knowledge, confidence, and perceived competence.

Comparing the pre-class and post-class results, there are statistically significant improvements in the knowledge, confidence and perceived competence scores amongst the graduate students surveyed. Our findings suggest the educational effectiveness of our AI-assisted biomarker review tutorial. The proposed workflow provides a practical approach for equipping students with the skills to use AI tools effectively and responsibly in systematic review workflows. Future work will focus on expanding the tutorial into a semester-long course and adapting it for other learner populations.

4. Becoming a Chemistry Teacher in the Age of AI: Pedagogical Encounters from School-Based Teaching Practice

Almubarak

(Online Presenter)

University of Pendidikan Indonesia

Abstract: Pre-service teachers are essential agents in educational transformation, especially in the context of artificial intelligence (AI) integration. Understanding student perceptions is key to assessing how AI can be meaningfully integrated into learning, especially regarding the level of trust and ease of use. However, studies exploring the use of AI by pre-service teachers in the context of teaching practice in schools are still limited. This study aimed to investigate students' experiences using AI in preparing and implementing learning during practice in schools. The method used was mixed methods with a sample of final semester chemistry education students at the University of Lambung Mangkurat, Indonesia, who were undergoing a teaching practice program in schools. The study refers to the Technology Acceptance Model (TAM) framework for the analysis of two main factors, namely “Perceived Usefulness” (PU) and “perceived ease of use (PEU).” Data were collected using questionnaires (quantitative) and in-depth interviews (qualitative). The results showed that in the PU dimension, around 82% of students felt that AI helped them understand complex chemistry concepts and helped them prepare for learning (86%). For the indicator that AI helps students find references and are ready to use it in learning, the percentage is the same at 96%, motivation (94%), and creativity in designing learning (96%). In the PEU dimension, 76% feel at ease regarding the accuracy of AI information, problem-solving skills (92%), and independent learning (76%), and all students agree that training/workshops on its integration in lectures are carried out.

5. Artificial Intelligence in An Engaging Science Classroom: A Standpoint of Prospective Professional Teachers in Madura, Indonesia

Aida Fikriyah

(Online Presenter)

University of Trunodjoyo Madura, Indonesia

Abstract: The rapid development of artificial intelligence (AI) has progressively become integrated into science education, restructuring traditional classrooms toward adaptive and inquiry-based environments, offering new role as a cognitive partner, and reinforcing the learning sources in class. In spite of the fact that, AI integration in education also provides impacts on students' learning achievement and teachers' dependence on technology and accessible learning. This study aims at exploring the pedagogical standpoint of science education students as prospective professional teachers on the implementation of AI in science classroom, specifically on its advantages, disadvantages, and challenges. A quantitative study using a survey with standardized questions was conducted toward students of teacher professional education program in a certain university. Findings indicate that students as prospective professional teachers strongly utilize AI tools in science classroom to assist in learning preparation, decision-making on task completion, implementation, evaluation, and assessment. Therefore, it can be concluded that students in teacher professional education program who are expected as prospective professional teachers completely claim that AI brings beneficial roles in science education, despite the fact that AI also impacts on teachers' dependence.

Oral Presentation Parallel Session 1C: AI, Human Experience, and Society

Session Number: 1C

Time: 12:30 to 14:30

Venue: TR703

1. The Use and Interpretation of Popular Memes (Geng) on the Chinese Short-Video Platform Douyin: A Qualitative Sociolinguistic Study of Chinese Short-Video Culture

Ki Kwong Kenrich Chen

(Online Presenter)

Stanford Pre-Collegiate Summer Institute

Abstract: This study examines the use and interpretation of Geng, a type of popular meme or joke, on the Chinese short-video platform Douyin. Drawing on a qualitative multiple-case study of 20 videos and related user comments, the analysis identifies a spectrum of Geng types ranging from low-context sensory humor (e.g., the “Jet2 Holiday” audio) to high-context cultural performances (e.g., the “American bullying” trope). The findings show patterned manipulation of language and visual elements through “translationese,” stylization, and multimodal resources. These recurring patterns can be interpreted as evidence of metapragmatic awareness. The discussion considers three possible social functions of these Geng practices: supporting shared recognition among users, framing uncomfortable content as playful and less realistic, and reframing everyday mishaps through humor. These functions illustrate how recurring Geng forms can support lightweight affiliation and playful participation in the sampled videos and comments.

2. AI-Mediated Education and the Reconfiguration of Human – World Relation

Chung-En Cheng

(Online Presenter)

Institute of Education, University of Strathclyde

Abstract: Generative Artificial Intelligence (Gen AI) is increasingly embedded in daily life; its impact extends beyond deliberate use. AI has become a medium shaping how humans perceive, interpret, and respond to the world; its influence is not just on a single answer to a question, but on how humans access and understand the world. AI's mediation should not be understood as creating an instrumental orientation from nothing. Rather, it may contribute to the intensification of an already existing tendency to approach and evaluate the world primarily in terms of efficiency, immediacy, and goal-attainment. In this context, the educational concern is not efficiency or instrumentality per se, but whether AI mediation reduces the plurality of relations through which humans encounter, interpret, and participate in the world.

Comparatively, little work has examined how the embedding of generative AI reconfigures the human–world relation. Accordingly, this paper examines how Gen AI reconfigures the human–world relation and argues that this reconfigured relation is best understood as alienation. Furthermore, it examines the implications for how education should respond to alienation. The examination will be based on Hartmut Rosa's (2021) concept of resonance and alienation to diagnose whether the reconfigured relation is alienation and illustrate how alienation can be seen as a pedagogical condition.

Philosophers of technology have long argued that technologies are more than mere tools, but rather forms of mediation (Verbeek, 2005). As one of the most influential technologies of the contemporary era, AI therefore needs to be examined not only in terms of how it may be used effectively, but also in terms of how it reshapes the conditions under which humans encounter the world. Thus, this paper draws on Rosa's distinction between Resonance and Alienation to evaluate whether the AI-mediated relation remains open to responsiveness. It suggests that alienation should be understood not only as a problem education must overcome, but also as a condition from which education may begin.

3. Human-AI Communication and Misinformation: Trust and Accuracy

Sumaiya Afreen Faria

(Online Presenter)

University of Dhaka

Abstract: As millions of people around the world rely on AI, understanding how trust and accuracy function in human-AI communication has become a scholarly concern among researchers. This research explores interdisciplinary connection on the trust, accuracy, and misinformation in human-AI interaction. This study addresses that gap by investigating the following research questions: (1) How do people judge the truthfulness of AI-generated information based on its wording?, (2) How do people respond in conversation to inaccurate AI information? By analyzing scholarship across the computer science, information, linguistic, and communication sciences to understand and evaluate user responses to AI outputs. Computer Science research tends to emphasize technical solutions, while Communication Studies research focuses on conversational behavior. Information Science looks at how people check facts and search for reliable information, while Linguistics looks at how word choice shapes whether people believe something is true. Findings reveal a shared gap across disciplines in trust and accuracy, as trust in AI is mostly studied within computational disciplines, existing research frequently measures human trust separately from actual system accuracy. This will be helpful to establish a foundational framework for future interdisciplinary research integrating computational, communicative, informational, and linguistic models to mitigate AI-driven misinformation by studying trust and accuracy, while providing important insights for software design, information literacy, and technology development.

4. History in Digital Hyperreality: An Analysis of AI Simulacra in TikTok Content on @hai.dakwah

Muhammad Azmi Favlevi

Universitas Islam Negeri Syarif Hidayatullah Jakarta

Abstract: The emergence of artificial intelligence (AI) has transformed the ways historical knowledge is reconstructed, represented, and consumed in digital media. On TikTok, historical learning is increasingly presented through short-form audiovisual content that combines narrative storytelling with AI-generated images, characters, voices, and visual environments. While this technological development provides new opportunities to make history more accessible, engaging, and relevant to younger audiences, it simultaneously raises questions about the relationship between historical reality and digitally constructed representation. This study examines how AI-generated historical learning content on the TikTok account @hai.dakwah constructs simulacra and produces forms of hyperreality through the perspective of Jean Baudrillard. The study focuses on how historical events, figures, and contexts are reconstructed through AI and subsequently presented as seemingly realistic digital representations. Employing a qualitative interpretive approach, the study analyzes selected AI-generated TikTok videos by examining their visual representations, narrative structures, characterization, digital environments, and relationship with historical references. Baudrillard's concepts of simulacra and hyperreality are used to interpret how technological reproduction may transform historical references into representations that acquire meaning independently from the original reality. The analysis indicates that AI enables historical narratives to be transformed into immersive visual experiences that can make past events appear immediate and tangible to contemporary audiences. However, the realism produced by AI may also obscure the distinction between documented historical reality and digitally constructed imagination. In this context, the AI-generated image does not merely function as an illustration of history but may become a new reference through which audiences encounter and interpret the past. The study highlights the pedagogical potential of AI-generated historical content while emphasizing the importance of critical digital literacy, particularly in distinguishing historical evidence from technologically produced representations.

5. The Associations of Environmental Identity, Collective Efficacy, and Social Norms with Climate Actions Among University Students

Kumarasan S/O Vijayakumar

Psychology and Child and Human Development, National Institute of Education

Abstract: Climate change is one of the most pressing global challenges of the 21st century, and understanding the psychological factors associated with climate action is critical to addressing it. Environmental identity, social norms, and collective efficacy have each been associated with climate action, but their relative associations with different types of climate action remain unclear. This study examined whether these factors were differentially associated with personal, social, and civic climate action.

Most previous research has focus on Western contexts, overlooking Asian populations such as Singapore, where this study was conducted. Although climate action has been examined across different types, such as personal, social, and civic action, few studies have compared whether environmental identity, social norms and collective efficacy show stronger associates with specific types of climate action to which they are theoretically most relevant.

Cross-sectional data was collected from 1174 Singaporean undergraduates aged 18-35 via Qualtrics. Three multiple linear regression analyses were conducted in R to examine the associations of environmental identity, social norms, and collective efficacy with personal, social, and civic climate action.

Environmental Identity showed the strongest association with personal action, whereas social norms showed the strongest associations with social and civic action.

Environmental identity was positively associated with personal and social action, while social norms were positively associated with all three types of climate action. In contrast, collective efficacy was positively associated with personal action but negatively associated with social and civic action.

The findings suggest that the psychological factors associated with climate action may differ across types of action, with Social norms relevance to social and civic action, and environmental identity relevance to personal action. Collective efficacy's negative associations with social and civic action suggest it may be less important for cultivating these forms of climate action, though it remains relevant for personal action.

Oral Presentation Parallel Session 1D: STEM Education and Computational Thinking

Session Number: 1D

Time: 12:30 to 14:30

Venue: TR704

1. Predicting Student Dropout Using Random Forest: An Early Warning System Approach

Lulu Ulrojabiah¹ (*presenter*), Sarini Abdullah²

Universitas Islam Negeri Syarif Hidayatullah Jakarta¹, Faculty of Mathematics and Natural Sciences (FMIPA), Universitas Indonesia²

Abstract: Student attrition remains a critical challenge within global higher education, necessitating proactive pedagogical interventions. While existing literature extensively explores traditional predictive models for student retention, these models frequently struggle with severe class imbalance and often act as "black boxes," overlooking the need for transparent interpretability in educational data. Thus, this study addresses two primary research questions: (1) How can an integrated machine learning framework accurately predict student dropout while handling imbalanced data? and (2) Which early multidimensional factors predominantly drive student retention? To answer these, this study evaluates a comprehensive higher education dataset comprising 4,424 student records, encompassing demographic backgrounds, socioeconomic status, and semester-wise academic performance metrics. This study proposes a robust early warning system utilizing a Random Forest classifier. To address the inherent class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) is applied exclusively to the training phase to prevent data leakage. Furthermore, SHapley Additive exPlanations (SHAP) is integrated to decode the model's decision-making process. The proposed framework demonstrates high predictive performance, achieving an overall accuracy of 78% and a robust ROC-AUC score exceeding 0.85. Empirical findings, supported by global SHAP interpretations, elucidate that first-year academic engagement, specifically the number of approved curricular units and semester grades are the most dominant predictors of student retention, significantly outweighing demographic and socioeconomic variables. By operationalizing these insights through transparent machine learning, educational institutions can dynamically identify at-risk students and understand their specific risk drivers, facilitating targeted academic advising that improves overall student success rates.

2. The Conceptualization and Identification of Epistemic Risk-Taking in Stem Problem Solving

Erti Hamimi (*presenter*), Jina Chang, Tan Aik Ling

Natural Sciences and Science Education, National Institute of Education

Abstract: STEM problem solving often involves uncertainty, unexpected challenges, and revision of ideas and solutions. Engaging productively in these moments may require students to take epistemic risks in developing knowledge. However, risk-taking has largely been examined as an individual characteristic, with limited attention to Epistemic Risk-Taking (ERT) in STEM problem solving. To address this gap, this qualitative study conceptualizes ERT through a review of the existing literature and examines how ERT is enacted during STEM problem-solving activities. Qualitative data consisted of lesson observations of three STEM lessons conducted at the Applied Learning Programme (ALP) secondary school in Singapore, an interview with the teacher, and a students' focus group. The data were analysed using discourse analysis to identify instances of ERT as they emerged through students' interactions and problem-solving processes. The findings illustrate the emergence of ERT during STEM problem-solving and provide empirical evidence of how students engage in risk-taking when constructing and developing knowledge. These findings contribute to the conceptualization of ERT and highlight its potential role in supporting productive engagement with uncertainty and knowledge construction in STEM learning.

3. Unlocking Creative Stem Ecosystems in Indonesian Secondary Schools during the Post-Truth Era: A Fuzzy Logic and Qualitative Analysis

Maya Umi Hajar

(Online Presenter)

Universitas Negeri Malang

Abstract: Implementation of STEM education in Indonesian secondary schools continues to face persistent challenges, particularly uneven teacher quality and disparities in students' socioeconomic support, both of which contribute to significant variation in learning quality across schools. In the current post-truth era, where evidence is frequently contested and public trust in expertise is increasingly fragile, education policy urgently needs to be grounded in transparent, verifiable, and data-driven models rather than intuition alone. This study addresses that need by developing a creative STEM ecosystem model based on a Fuzzy Inference System (FIS), aimed at identifying the dominant factors shaping STEM implementation at the secondary school level. Using a sequential mixed-methods design, quantitative data were collected from 85 teachers and 68 parents and analyzed through fuzzy inference modeling, complemented by in-depth interviews with key education stakeholders to enrich and contextualize the quantitative findings. The results indicate that teachers' cognitive capacity is the most influential determinant of STEM implementation, followed by students' socioeconomic background, with the resulting model demonstrating strong predictive validity ($R^2 > 0.95$). These findings provide a robust empirical foundation for formulating evidence-based education policy that strengthens teacher capacity as a strategic entry point for building sustainable STEM ecosystems. By combining computational modeling with qualitative insight, this study contributes methodologically through the novel application of fuzzy logic in secondary-level STEM ecosystem research, and conceptually by repositioning teacher cognitive capacity as a central rather than peripheral factor in STEM ecosystem formation. The study supports the achievement of Sustainable Development Goal 4 (Quality Education) by offering a replicable, evidence-based framework for strengthening STEM education practice in developing-country contexts.

4. High School Students Perceptions, Knowledge, Attitudes, and Behaviors Toward Microorganism Issues: A Correlational Study

Ali Mustofa

(Online Presenter)

Universitas PGRI Ronggolawe

Abstract:

Purpose – This study aimed to examine the structural relationships among high school students' perceptions, knowledge, attitudes, and hygienic behaviors toward microorganisms within the Knowledge-Attitude-Practice (KAP) framework. The study addresses the persistent gap between students' microbiological knowledge and its application in everyday health practices.

Methodology – A quantitative cross-sectional survey was conducted involving 188 tenth- and eleventh-grade students from a senior high school in Tuban Regency, East Java, Indonesia, selected through purposive sampling. Data were collected using instruments measuring perceptions, knowledge, attitudes, and hygienic behaviors. The hypothesized relationships were analyzed using Partial Least Squares Structural Equation Modeling (PLS-SEM) with 1,000 bootstrap subsamples.

Findings – Most students demonstrated good knowledge (78.2%) and positive attitudes (70.2%), whereas consistent hygienic behaviors were reported by only 58.6%. PLS-SEM showed that perceptions significantly predicted knowledge ($\beta = 0.308$, $p < 0.001$), knowledge predicted attitudes ($\beta = 0.187$, $p = 0.011$), and attitudes predicted hygienic behaviors ($\beta = 0.251$, $p = 0.001$). In contrast, perceptions did not significantly predict attitudes, and knowledge did not directly predict behaviors.

Novelty – The study empirically demonstrates a sequential pathway from perceptions to knowledge, attitudes, and hygienic behaviors, extending the application of the KAP framework to microorganism-related learning in secondary education.

Significance – The findings indicate that microbiology education should move beyond knowledge acquisition by integrating cognitive development, attitude formation, and behavioral habituation to facilitate the translation of learning into everyday health practices.

5. When Machines Need Trained Eyes: A Curriculum Framework for Weather Observer Education in the Age of AI

Joan E. Maghacot

Lyceum Northwestern University

Abstract: Artificial intelligence (AI), satellite technologies, automated instruments, and networked sensors are transforming National Meteorological and Hydrological Services (NMHS), requiring weather observer education to evolve alongside technological advancement. Despite growing research on AI-enabled workforce development, limited attention has focused on meteorological observer training, where human expertise remains critical for validating automated data, detecting anomalies, and contextualizing AI-supported forecasts. This study examines the competencies required for effective human–AI collaboration in meteorological observation and explores how academia, technology, and operational institutions can collaboratively redesign professional training. Using comparative case analysis, two illustrative operational incidents are examined: detection of an erroneous instrument reading caused by water ingress and visual identification of a typhoon feeder band before satellite detection. Grounded in competency-based education and adult-learning theory, the analysis identifies teachable competency gaps and proposes a four-domain curriculum framework encompassing contextual visual observation, instrument quality-assurance diagnosis, extreme-event operational continuity, and structured mentorship for transferring tacit professional knowledge. Rather than positioning AI as a replacement for human expertise, the framework advances a human-in-the-loop approach that integrates technological capabilities with professional judgment and situational awareness. The study provides NMHS training institutions and academic partners with an adaptive, evidence-based model for strengthening workforce readiness while bridging education, technology, and operational practice in the age of AI.

Oral Presentation Parallel Session 1E: Environment and Sustainability Education

Session Number: 1E

Time: 12:30 to 14:30

Venue: TR705

1. The 2025 Senyar storm: Malacca Vortex as an emerging hazard over Southeast Asia?

Dong Luojie

Natural Sciences and Science Education, National Institute of Education

Abstract: In November 2025, an unprecedentedly intense vortex named Senyar developed over the Malacca Strait, a location typically unfavorable for severe storm intensification. Shallow straits promote upwelling and sea surface cooling, narrow fetch limits energy supply, and weak low-latitude Coriolis force constrains vorticity. Nevertheless, Senyar formed under these adverse conditions, bringing extreme rainfall, damaging winds, and landslides that caused over 1000 fatalities across southern Thailand, the Malay Peninsula, and northern Sumatra. This event ranks among the most extreme precipitation events in recorded history. Planetary-scale diagnosis reveals that constructive interactions between an equatorial Rossby wave and an active Madden-Julian Oscillation (MJO) phase provided a favorable large-scale envelope, while a concurrent negative Indian Ocean Dipole (IOD) supplied sustained moist energy. At the synoptic scale, the anomalously lower mid-tropospheric geopotential heights and persistent low-level moisture convergence triggered intense deep convection with high cloud tops, leading to a sustained low-pressure system. A clearly defined mesoscale vortex emerged within the broader convective envelope, exhibiting a compact circulation center and organized spiral rainbands. The system crossed the Malacca Strait twice: first after crossing the Malay Peninsula and making landfall in northeastern Sumatra, and again after re-emerging over the strait prior to its second landfall in Selangor. In terms of predictability, TIGGE ensemble members reveal systematic forecast biases: extreme precipitation was widely underestimated, and the simulated low-pressure center exhibited substantial track errors. These findings underscore the challenge of predicting topographically constrained tropical cyclogenesis in historically immune regions like the Malacca Strait - a threat that may grow under climate warming, demanding heightened vigilance and improved forecasting.

2. Artificial Intelligence for Flood Prediction: What Works, Where, and Why?

Huang Jiachun

Humanities and Social Studies Education, National Institute of Education

Abstract: Artificial intelligence is increasingly used to predict floods, river discharge, and inundation. However, rapid growth in AI-based hydrology has produced a wide range of models, datasets, evaluation methods, and reported performance, making it difficult to determine when AI provides a real advantage over conventional approaches. This systematic review examines how AI and machine learning are being used for river and flood assessment. It focuses on four questions: What hydrological problems are AI models used to solve? What data and model types are commonly used? How are models evaluated? Under what conditions does AI improve prediction? Studies will be classified by application, including rainfall–runoff modelling, streamflow prediction, flood forecasting, flood mapping, and extreme-event assessment. Particular attention will be given to commonly used approaches such as neural networks, recurrent and transformer-based models, hybrid process–AI models, and emerging Earth-system foundation models. The review will compare model accuracy, forecast lead time, data requirements, interpretability, spatial transferability, and performance during extreme floods. It will also identify major limitations, including limited observations, inconsistent validation, poor transfer between basins, and uncertainty under changing climate conditions. The aim is to provide a clear framework showing where AI is useful, where conventional models remain necessary, and how the two can be combined for more reliable flood prediction.

3. Designing an Investigative Atmospheric Science Course for Climate Change Awareness

Tika Elviana

(Online Presenter)

Universitas Pendidikan Indonesia (Indonesia University of Education)

Abstract: Climate change awareness is increasingly important in higher education, particularly for students studying atmospheric processes, weather, and climate. Investigative science learning emphasizes active engagement in observing phenomena, examining evidence, and constructing scientific explanations (Etkina & Planinsic, 2024), while Evidence-Based Learning highlights the use of evidence to support learning and knowledge construction (Rajaram, 2021). However, atmospheric science courses are still often dominated by theoretical explanations, providing students with limited opportunities to investigate atmospheric phenomena and connect scientific concepts with observable environmental conditions. This study aims to design an investigative atmospheric science course that supports undergraduate students' climate change awareness. The study addresses the question: How can an atmospheric science course be designed to engage students in investigating weather and climate phenomena and relating their findings to climate change? The study follows an instructional design approach involving analysis of course learning outcomes, relevant literature, atmospheric science content, and appropriate investigative learning activities. The proposed course design organizes learning around observing atmospheric phenomena, formulating questions, conducting investigations, examining evidence, discussing findings, and constructing scientific explanations. Topics include atmospheric processes, weather elements, interactions among atmospheric variables, and their relationships with climate and climate change. The expected outcome is an instructional framework that provides students with more active and meaningful opportunities to connect atmospheric science concepts, empirical evidence, and climate change issues. The framework may provide a practical basis for integrating investigative learning into undergraduate atmospheric science education.

4. Eco-Residential Design Inspired by Lantiok Malay-Riau Traditional House as a Learning Resources: Evidence from Thermal and Airflow Visualization Using Computational Fluid Dynamics (CFD)

Diniya

(Online Presenter)

Universitas Pendidikan Indonesia

Abstract: Vernacular architecture represents not only cultural heritage but also empirical knowledge developed through long-term adaptation to local environments. The Lontiok house, a traditional Malay dwelling in Kampar, Riau, incorporates climate-responsive features such as building orientation, elevated construction, spatial connectivity, and strategically positioned openings that influence natural ventilation and indoor thermal conditions. This study investigates the fluid-dynamics principles embedded in Lontiok architecture, translates these principles into contemporary eco-residential designs, and visualizes airflow phenomena using Computational Fluid Dynamics (CFD). An exploratory case-study approach was employed, integrating literature review, field observation, architectural design development, and ANSYS CFD simulation. Findings from four purposively selected articles and direct observation were synthesized to identify cultural, architectural, and environmental characteristics relevant to airflow and thermal performance. These principles were then reinterpreted into four modern residential layout alternatives. CFD simulations showed that opening configuration, interior partitions, and spatial connectivity strongly influenced airflow continuity, recirculation zones, and temperature distribution. Although all layouts supported natural ventilation, the fourth alternative produced the most continuous airflow path through major occupied spaces and a relatively more uniform thermal distribution, despite localized low-velocity and warmer zones. The results indicate that ventilation performance depends not only on maximum airflow velocity but also on airflow continuity and spatial distribution. CFD visualization further demonstrates the educational potential of Lontiok architecture by making otherwise invisible phenomena, including airflow, pressure-driven ventilation, recirculation, and temperature gradients, observable. The study provides a proof of concept for using vernacular architecture as a contextual resource linking local heritage, architectural innovation, and fluid-dynamics education.

5. Advancing AI-Driven Discipline-Centric, Agile, and Sustainable Innovation Growth (DASIG) Framework for Global University Sustainability

Daniel D. Dasig Jr

Research and Development Office, Philippine Women's University

Abstract: Global university ranking systems, including QS, Times Higher Education (THE), and the World University Rankings for Innovation (WURI), increasingly influence institutional reputation, competitiveness, internationalization, and strategic development. However, universities face the challenge of responding to global performance indicators while preserving their fundamental missions in teaching, research, community engagement, and sustainable development. This study advances an AI-Driven Discipline-Centric, Agile, and Sustainable Innovation Growth Framework for strengthening global university sustainability. The framework integrates discipline-specific strengths, organizational agility, sustainable innovation, AI-enabled smart workflows, and intelligent assistants to enhance research productivity, educational innovation, graduate employability, industry collaboration, social impact, and evidence-informed decision-making. Using case-based analysis and institutional performance data, the study examines how AI-supported processes can identify performance gaps, streamline academic and administrative workflows, optimize resource allocation, and strengthen institutional responsiveness to evolving global higher education indicators. The framework emphasizes responsible and human-centered AI adoption, ensuring that technological transformation reinforces rather than compromises academic priorities and institutional values. Findings demonstrate the potential of integrating AI, disciplinary excellence, agility, and sustainability to address ranking-related pressures while generating meaningful institutional outcomes. The framework offers universities an adaptive pathway for bridging academia, technology, and industry while advancing sustainable innovation, global competitiveness, institutional resilience, and long-term educational impact.

Oral Presentation Parallel Session 1F: Identity, Engagement, and Educational Development

Session Number: 1F

Time: 12:30 to 14:00

Venue: TR706

1. Mechanisms of Identity and Well-Being: The Mediating Role of Eudaimonic Orientation Across Identity Statuses

Koh Wei Xun

Psychology and Child and Human Development, National Institute of Education

Abstract: Identity development is a core aspect of psychosocial functioning and has been linked to various dimensions of well-being. However, the mechanisms through which identity influences well-being remain underexplored. This study sought to examine the mediating role of well-being orientations in the relationships between identity statuses and well-being outcomes. The study was conducted in Singapore and the sample consisted of 267 adults ($M = 28.41$, $SD = 5.33$). A cluster mediational approach that combined person and process-oriented methods was utilized. Cluster analysis replicated all four of Marcia's original identity status clusters – (1) Achievement (2) Foreclosure (3) Moratorium (4) Diffusion. Follow-up ANOVA and post-hoc tests highlighted significant well-being differences between the identity achievement and diffusion clusters. Mediational analysis showed that eudaimonic orientation mediated the pathways between identity status clusters and both hedonic and eudaimonic well-being outcomes. Hedonic orientation on the other hand was not associated with most of the identity statuses and well-being outcomes. The results and discussion suggest that eudaimonic orientation evolves with identity status changes and contributes to well-being in tandem with identity development. Future studies should incorporate longitudinal panel data to model identity growth trajectories to further investigate this phenomenon.

2. Embracing the global without leaving home: a collective case study of “internationalization at home” in Chinese higher educational institutions from the faculty perspective

Zou Zheng

Humanities and Social Studies Education, National Institute of Education

Abstract: Traditionally, cross-border mobility has been regarded as a key indicator of the Internationalization of higher education, especially the mobility of academic personnel, such as students studying abroad. The unexpected pandemic has reshaped virtually every aspect of global life, with higher education being no exception. The pandemic's most immediate impact was the disruption of international student mobility. In light of these trends, the concept of “internationalization at home” (IaH) has become popular. The focus of IaH is on the absence of physical cross-border mobility.

Faculty, as principal agents in curriculum design, instruction, and research, have a substantial impact on how internationalization is integrated into the educational experience. They play a central role in both implementing and disseminating internationalization in line with institutional or national strategies. Therefore, understanding how faculty conceptualize and engage with IaH is an essential first step in advancing IaH in China.

3. Bridging Academic Knowledge and Professional Practice through the Practicum: Exploring a Preservice Mathematics Teacher's Noticing

José L. Morales-Reyes

Pontifical Catholic University of Valparaíso

Abstract: The practicum is a critical context in initial teacher education where preservice teachers encounter the complexities and constraints of classroom practice, apply and contrast theoretically knowledge, and develop situated and reflective professional competencies. Among these competencies, noticing enables teachers to make sense of significant classroom events and students' mathematical thinking. A recent conceptualization characterizes noticing through three interrelated dimensions: attending, interpreting, and shaping, with the latter referring to practices through which teachers shape classroom interactions to gain further access to students' thinking. However, evidence remains limited on how these dimensions manifest in practice, particularly among preservice teachers during the practicum. This instrumental case study addresses the question: How does teacher noticing manifest in a preservice mathematics teacher while teaching fractions during the practicum? The participant was completing his final practicum with seventh-grade students (ages 13–14). Data consisted of weekly written narratives about his own classroom experiences teaching fractions, produced using research-informed writing guidelines and followed by feedback from a university tutor. Findings show sustained attention to students' mathematical thinking across the narratives. His interpretations mobilized pedagogical content knowledge with varying depth, including recognition of students' errors and difficulties, connections with previous and prospective learning goals, extension of whole-number properties to rational numbers, fraction representations, and multiplication as repeated addition. The most evident changes occurred in shaping: eliciting details remained recurrent, whereas greater variation emerged in leveraging participation structures and creating and using resources. By the final narrative, monitoring different student productions, incorporating multiple voices, and using students' ideas as resources became coordinated around a shared mathematical difficulty. The study illustrates how research-informed narrative writing and university tutor feedback within the practicum can create opportunities to connect academic teacher education with the situated complexity of professional classroom practice.

4. Emerging Patterns of Generative AI Dependency Among Indonesian High School Students: A Cross-Sectional Study

Jasmine Nurul Izza

Universitas Negeri Malang

Abstract: The rapid integration of generative artificial intelligence (GenAI) into education has raised concerns about students' increasing reliance on AI-assisted learning. Drawing on Self-Determination Theory and Goodman's behavioral addiction framework, this study conceptualizes GenAI dependency as a multidimensional phenomenon reflected in cognitive preoccupation, negative consequences, and withdrawal-related symptoms. Despite the growing use of GenAI among adolescents, empirical evidence regarding these dependency patterns remains limited, particularly in the Indonesian secondary education context. This cross-sectional study investigated patterns of GenAI dependency and gender differences among Indonesian senior high school students. Data were collected through an online questionnaire from 507 students across Java and other Indonesian islands and analyzed using descriptive and inferential statistics. The findings revealed distinct patterns across the three dimensions. Cognitive preoccupation was generally moderate, suggesting that GenAI had become cognitively salient for some students without necessarily dominating their learning activities. Negative consequences were comparatively limited, indicating that most students had not yet experienced substantial adverse effects associated with GenAI use. In contrast, withdrawal-related symptoms showed the highest level of endorsement among the measured dimensions, suggesting that discomfort or difficulty when access to GenAI is restricted may emerge before more observable negative consequences. No significant gender differences were identified in GenAI dependency. Overall, the findings suggest an emerging rather than severe pattern of dependency, characterized particularly by withdrawal-related responses while substantial negative consequences remain limited. These findings highlight the importance of strengthening students' AI literacy and promoting autonomous and self-regulated engagement with GenAI before problematic patterns become more established.

5. When Does Anthropomorphism Help or Harm Science Learning? A Systematic Review of Its Implementation, Conditional Effects, and Calibration in Science Classrooms

Muhamad Wafda Jamil

(Online Presenter)

Seoul National University

Abstract: Science and science education have traditionally advised avoiding anthropomorphism. Anthropomorphic reasoning has often been linked to unscientific, inaccurate, or misleading ideas that may persist if they are not addressed. Yet anthropomorphism can help teachers explain unfamiliar science concepts. This tension leaves science instructors, especially teachers, unsure whether to use or avoid it. This systematic review examines evidence on the use of anthropomorphism in science classrooms from 1977 to 2025. We identified 51 empirical studies across science subjects and educational levels, from primary school to higher education. The review examines (i) the forms and instructional contexts in which anthropomorphism occurs in formal science education; (ii) the learning-related outcomes reported across educational settings; and (iii) the challenges, contextual conditions, and strategies associated with its use. The findings show that anthropomorphism is common and difficult to avoid in science teaching. Its influence should therefore be understood as conditional and context dependent, rather than inherently helpful or harmful. Complete avoidance may make science less accessible to learners. Teachers play a central role in calibrating anthropomorphism and helping students move from anthropomorphic language toward scientific reasoning. Based on these findings, the review proposes principles for the responsible use of anthropomorphism in science learning.

Oral Presentation Parallel Session 1G: Tamil

Language, Literature, and Education

Session Number: 1G
Time: 12:30 to 14:30
Venue: TR708

1. Beyond the Classroom: Reimagining Tamil Language Education through Communities of Practice (வகுப்பறைக்கு அப்பால் - சமூக நடைமுறைகள்வழி தமிழ்மொழிக் கல்வியை மறு நிர்மாணித்தல்)

(Presentation in Tamil)

Narayanan S/O Velayutham

Asian Languages and Cultures, National Institute of Education

Abstract: The rapid advancement of artificial intelligence, digital technologies, and evolving language practices has transformed the landscape of Tamil language teaching and learning. While considerable attention has traditionally been devoted to formal school education and teacher development, comparatively little focus has been placed on the continuing education of non-teaching Tamil language professionals. Translators, interpreters, media practitioners, librarians, cultural practitioners, public communications professionals, and language specialists play a pivotal role in shaping the contemporary use, visibility, and vitality of Tamil beyond the classroom. However, opportunities for structured professional learning, interdisciplinary collaboration, and systematic knowledge exchange remain limited.

This paper examines the emerging challenges faced by these professionals, including rapid technological disruption, the integration of AI into language work, fragmented professional development pathways, the transfer of tacit knowledge across generations, and the need to balance linguistic authenticity with evolving communicative demands. At the same time, it highlights the opportunities presented by lifelong learning and collaborative professional development to strengthen the Tamil language ecosystem.

Drawing upon Wenger's theory of Communities of Practice (CoP), the paper proposes a pedagogical framework that positions learning as a social and practice-based process. Rather than viewing professional development as isolated training events, Communities of Practice enable professionals across different sectors to collaboratively construct knowledge, share experiences, mentor emerging practitioners, and develop collective solutions to common challenges. Such communities foster reflective practice, interdisciplinary collaboration, innovation, and continuous capability development while strengthening professional identity and sustaining institutional knowledge.

The paper argues that Tamil language education must extend beyond formal educational institutions to embrace a broader ecosystem of lifelong professional learning. Establishing cross-sector Communities of Practice, supported by mentoring, experiential learning, digital collaboration platforms, and micro-credentialing opportunities, can create a resilient and future-ready Tamil language workforce. As language professions continue to evolve alongside technological innovation, collaborative learning models offer a sustainable pathway to enhance professional competencies while safeguarding linguistic quality, cultural authenticity, and the long-term vitality of Tamil in Singapore's multilingual society.

This abstract is appropriate for an academic conference and positions Communities of Practice not merely as a professional development model, but as a pedagogical approach to lifelong Tamil language learning for non-teaching professionals. It also aligns well with current discussions on AI, workforce development, and language sustainability.

2. Singapore in Tamil Literature: Society, History and Change

(தமிழ் இலக்கியத்தில் சிங்கப்பூர்: சமூகம், வரலாறு, மாற்றம்)

(Presentation in Tamil)

Saraswathi D/O Valliappan

Asian Languages and Cultures, National Institute of Education

Abstract: Identity and address proofs, legal documents, land records, and historical inscriptions form the foundational archives that chronicle a nation's legal, administrative, and cultural heritage. Beyond these official records, however, literature emerges as a profound and dynamic mirror of a society's pulse.

This article examines how a country's literary traditions illustrate the intersecting social, economic, and political lives of its people, and why citizens must engage deeply with them to help propel the nation forward. At its core, the study is guided by the **"Socio-Political-Economic Integration"** theory—a composite framework explaining how political structures, economic activities, and social systems operate in mutual dependence (Schneider, 2017; Raihan, 2016). The literary works analyzed here serve as practical demonstrations of this integrated framework in action.

Centering specifically on Singapore, this research explores how the nation's history and contemporary societal shifts have been preserved through local Tamil literature. By closely examining a selected novel, a short story, and a poem by Singaporean authors, this article paints a vivid picture of past ways of life, enduring customs, and the evolving structure of society.

3. Learning Tamil Through Play: Purposeful Play Pedagogy in the Primary Classroom

(விளையாட்டின்வழித் தமிழ் கற்றல்: தொடக்கநிலை வகுப்பறையில் விளையாட்டுவழி தமிழை சுவாரசியமாகக் கற்பித்தல்)

(Presentation in Tamil)

Kayal Vili d/o Pugalendi

Asian Languages and Cultures, National Institute of Education

Abstract: Play Pedagogy provides meaningful opportunities for students to learn and use Tamil in an engaging, interactive and purposeful manner. Rather than using games solely for enjoyment, purposeful play enables students to develop language skills through exploration, communication, collaboration and problem-solving.

This presentation will share practical ways of integrating Play Pedagogy into Tamil Language lessons across Lower and Upper Primary. For Lower Primary students, examples will include movement-based activities, language games, imaginative play and simple role-play to develop vocabulary, speaking and basic language structures. For Upper Primary students, more complex forms of play such as simulations, challenges, collaborative problem-solving and story-based activities will be explored to develop oral communication, vocabulary, grammar and writing skills.

The presentation will also highlight the important role of the teacher in designing purposeful play experiences that are aligned with clear learning outcomes and provide evidence of student learning.

Objectives

Participants will:

1. Understand how Play Pedagogy can support Tamil Language learning.
2. Explore practical play-based strategies for Lower and Upper Primary students.
3. Learn how play activities can be aligned with specific language-learning outcomes.
4. Gain ideas that can be adapted for their own Tamil Language classrooms.

Key Message

Play is not separate from learning. When purposefully designed, play creates meaningful opportunities for students to learn, communicate and use Tamil with confidence and enjoyment.

4. Evolution of Education System, Students and Teachers (மாற்றம் ஒன்றே மாறாதது - மாறிவரும் கல்விமுறையும் மாணவர்களும் ஆசிரியர்களும்)

(presentation in Tamil)

Shyamala Ganesan

Asian Languages and Cultures, National Institute of Education

Abstract: In this presentation, we will explore the updates implemented by the Ministry of Education (MOE) in the teaching-and-learning framework over the past five years, the current psychological state of students, the assistance available to novice teachers, the challenges new educators face in the classroom, and practical solutions to address these issues.

Furthermore, we will examine how the various educational theories we study are structured, how they benefit-or fail to benefit us, the indispensable role and consequences of Artificial Intelligence (AI) in today's educational landscape, and the crucial importance of maintaining positive teacher well-being.

Over the past five years, there have been massive and rapid shifts in teaching-and-learning methodologies as well as in the roles of teachers and students. This transformation is driven by the meteoric rise of generative AI and the Ministry of Education's progressive focus on prioritising student well-being. While using AI was initially viewed as a form of "cheating" today it has integrated into a core pillar of education. Though we have not fully transitioned to a stage where lessons cannot exist without AI, we must accept that it has become absolutely inevitable. Beyond physically attending school, learning through online mediums and continuous training through education apps has expanded significantly. Moving beyond the traditional whiteboard, methods like Virtual Reality and gamified learning have made immersive, experimental learning widely accessible today.

Teachers have transitioned from content deliverers who memorise and lecture lessons into facilitators who teach students how to critically evaluate which information is correct or false amidst the vast sea of data available on the internet and AI platforms, Educators have also begun utilising AI tools for curriculum design, lesson planning and automated grading. Concurrently, students are no longer just passive consumers of content, from a young age, they have developed the skills to autonomously build projects using AI applications. However, despite these advancements, current research indicates that student distractions have multiplied and the workload for teachers has actually intensified. In this presentation we will examine the solutions to these gaps and explore how teachers can uplift themselves both mentally and professionally.

5. Enhancing Picture Composition Writing through AI-Supported Collaborative Learning (செயற்கை நுண்ணறிவு - கூடிக் கற்றலின் மூலம் படக்கட்டுரை எழுதும் திறனை மேம்படுத்துதல்)

(Presentation in Tamil)

Ahuruachelam Shanthi

Asian Languages and Cultures, National Institute of Education

Abstract: Writing a good picture composition requires students to do more than simply describe what they see. Students need to observe closely, interpret situations, generate relevant ideas, organise events, and express their thoughts coherently. However, many Primary 4 and 5 students find it challenging to develop meaningful introductions and conclusions. They may rely on memorised sentences or surface-level descriptions instead of developing creative and relevant ideas based on the given pictures.

This presentation will share how Snorkl, an AI-supported learning platform, can be integrated into Tamil picture composition lessons to encourage deeper thinking, collaborative learning and greater student independence.

Students will first work collaboratively to observe the picture, discuss different possibilities and generate ideas for the introductions and conclusions. Carefully designed prompts will guide students to think about what may have happened before or after the scene, how the characters may be feeling, why events may have occurred and what evidence from the picture supports their ideas.

Snorkl will then be used as a form of digital scaffolding, allowing students to explain their ideas, receive timely feedback and refine their thinking before progressing to independent writing. The presentation will also highlight the important role of the teacher in designing meaningful prompts, facilitating discussion, differentiating support and ensuring that AI supports rather than replaces students' thinking.

Objectives

Participants will:

- Understand how AI-supported learning can strengthen thinking and idea generation in Tamil picture composition.
- Explore collaborative strategies for helping students develop meaningful introductions and conclusions. Learn how AI prompts and feedback can be used to support observation, inference, sequencing and reasoning.
- Gain practical ideas for differentiating support and guiding students towards more independent writing.

Key Message

AI should not give students the story. When purposefully used, it should help students think about the story, refine their ideas and become more confident and independent writers.

6. Enhancing Students' Speaking Skills and Short-Answered Responses through Artificial Intelligence (ClassPoint) (செயற்கை நுண்ணறிவு (Classpoint) வழியாக மாணவர்களின் பேசுதல் திறன் மற்றும் குறுகிய வினாக்களுக்கு விடையளித்தலை மேம்படுத்துதல்)

(Presentation in Tamil)

Chek Davoudou Raghmath Nissa

Asian Languages and Cultures, National Institute of Education

Abstract: Among students learning Tamil as a Second Language in Singapore schools, some come from non-Tamil-speaking backgrounds. These students have different levels of prior knowledge, interests, and learning styles. The Ministry of Education's core philosophy states: "Every student wishes to learn; every student can learn." Guided by this principle, this article explores ways to reduce learning gaps by differentiating subject content and classroom delivery to engage students effectively.

To support this goal, the article examines how the ClassPoint platform is used to create teaching resources, implement them in the classroom, monitor student progress, and address key learning challenges. By integrating ClassPoint into the teaching approach, students not only improve their speaking skills and ability to provide short-answer responses but also develop essential 21st-century skills.

Through ClassPoint, self-directed learning, which students practise through Singapore's Student Learning Space (SLS), can be effectively integrated into classroom learning. This encourages active participation and transforms learning into a more engaging and enjoyable experience (Joy of Learning). Furthermore, lesson materials tailored to students' learning needs and progress encourage greater independence, enhance speaking skills, and strengthen teacher-student interaction.

In summary, ClassPoint provides an integrated language-learning environment while creating a fun and engaging space for students to speak and write in Tamil, both within and beyond the classroom.

Poster Presentations

Venue: Outside LT1

1. International Chinese Language Teachers' In-the-Moment Professional Judgment in AI-Assisted Teaching Contexts: Underlying Rationales and Instructional Intervention Choices

Rao Ziyi

Asian Languages and Cultures, National Institute of Education

Abstract: This study tentatively explores teachers' in-the-moment professional judgment in AI-assisted Chinese language teaching. Existing literature on teacher cognition and pedagogical reasoning has emphasized that teaching is not merely the application of techniques, but a process of situated professional judgment. Shulman's discussion of pedagogical reasoning highlights teachers' capacity to understand, transform, and justify pedagogical decisions, while van Manen's notion of pedagogical tact draws attention to teachers' sensitive responses to students in concrete classroom moments. Meanwhile, recent studies on AI-assisted language teaching have mainly focused on the functions of AI tools, such as error correction, feedback, personalization, and learner support. However, less attention has been given to how language teachers make immediate judgments when using or working alongside AI, especially in the context of teaching Chinese as an international language.

The research gap, therefore, lies in the limited understanding of how Chinese language teachers diagnose students' surface problems, redefine teaching tasks, and adjust pedagogical intervention in real-time classroom situations. Based on eight reflective teaching cases, this study asks: In AI-assisted Chinese language teaching, how do teachers move from students' surface problems to deeper diagnosis, redefine the teaching task, and adjust pedagogical intervention according to classroom context?

Methodologically, this study will adopt a framework-building approach rather than large-scale empirical research. The existing eight cases will be re-analysed to identify recurring dimensions of teachers' professional judgment. Tentatively, the framework may include four dimensions: diagnostic judgment, task redefinition, contextual weighing, and pedagogical intervention calibration. The expected contribution is to propose an initial analytical framework that explains how teachers exercise situated professional judgment in AI-assisted Chinese language teaching. Rather than arguing that teachers are simply "irreplaceable" because AI cannot perform certain tasks, this study aims to show how teacher expertise is reflected in the capacity to interpret learners, tasks, and classroom contexts, and to make pedagogically appropriate interventions.

2. Development of an embodied learning simulation to support students' academic anxiety regulation

Youngsun Choo

Seoul National University

Abstract: Embodied learning emphasizes that learning emerges through interactions among the body, cognition, and environment. From this perspective, learning to regulate emotions requires learners to recognize and modify bodily and cognitive responses in relation to the situations in which emotions occur. Academic anxiety is particularly relevant for university students, who frequently encounter evaluative and challenging academic situations. However, limited research has examined how embodied learning can inform the design of learning environments for academic anxiety regulation. This study aimed to develop an embodied learning simulation that enables university students to repeatedly practice and embody strategies for regulating academic anxiety. Based on the literature on embodied learning and academic anxiety regulation, we identified bodily arousal, cognitive interpretation, and attentional regulation as key components. Academic anxiety was conceptualized as a process in which threatening predictions in challenging academic situations elicit bodily arousal, which may be interpreted as a threat and amplify anxiety, consequently disrupting attention to learning. Accordingly, three design principles were developed: (1) supporting awareness of bodily arousal, (2) supporting adaptive interpretation of bodily responses and academic situations, and (3) supporting attentional reorientation toward learning tasks. These principles informed the development of a simulation in which students repeatedly experience anxiety-provoking academic situations and practice regulation strategies. Using a design-based research approach, the simulation and design principles were iteratively developed and refined in collaboration with university students and counseling experts. The study examines how the proposed principles can be instantiated in an embodied learning simulation and refined through stakeholder evaluation, ultimately providing design knowledge for technology-supported academic anxiety regulation.

3. Exploring the Facilitative Role of Conversational AI in Collaborative Visual Arts Learning

Zhou Wenqian

Learning Sciences and Assessment, National Institute of Education

Abstract: Collaborative learning is widely used in visual arts education to foster creativity, reflection, and communication. However, challenges such as uneven participation, delayed feedback, and emotional fluctuations may undermine students' engagement and collaborative interaction. Conversational Artificial Intelligence (CAI), such as ChatGPT, has shown potential to support learning through adaptive feedback, dialogic scaffolding, and facilitation. Nevertheless, existing empirical research has largely focused on STEM, programming, and language learning, while the facilitative role of CAI in collaborative visual arts learning remains underexplored.

This proposed study investigates how CAI facilitation influences university students' engagement and their perceived quality of interaction during collaborative visual arts learning. It addresses three questions: (1) To what extent does CAI facilitation influence students' behavioral, emotional, and cognitive engagement? (2) How do students perceive CAI's role in shaping the quality of interaction during collaboration? and (3) What differences emerge between Group-CAI and Individual-CAI facilitation in sustaining engagement and perceived interaction quality?

A mixed-methods experimental design is proposed involving 24 undergraduate Visual Communication Design students working in six groups. Three facilitation conditions will be compared: Group-CAI, where one shared ChatGPT facilitates the group; Individual-CAI, where each student accesses ChatGPT individually; and a No-CAI control condition. Participants will undertake a four-week collaborative visual identity design project. Data will be collected through pre- and post-engagement questionnaires, reflection journals, and semi-structured interviews, and analyzed using quantitative and thematic approaches.

The study aims to provide empirical insights into how CAI can function as a facilitator in collaborative creative learning and inform the meaningful design of AI-supported collaborative pedagogy in visual arts education.

4. When AI Audits: What the Profession Demands, and What Audit Education Must Deliver

Jingya Li

Capital University of Economics and Business

Abstract:

Key Literature and Research Gaps

The rapid deployment of AI in audit practice has outpaced academic research on how entry-level auditors should be prepared. While prior studies have examined generic skill shifts in accounting, few have grounded their analysis in the actual language of employer demand. This study addresses this gap by using job advertisements as a direct lens through which to identify the competencies now prioritized by the profession.

Research Question

What competencies do Big Four firms signal as essential for entry-level auditors in an AI-augmented environment, and to what extent are these signals reflected in current university audit curricula?

Key Findings

An exploratory analysis of 15 entry-level audit job postings from Big Four firms, triangulated with a review of audit-related course titles at five Chinese universities, yields three insights. First, the profession's demand for human judgement is paramount: 80% of postings explicitly require critical thinking and professional scepticism, while only 47% require specific technical tools. This suggests that AI integration in audit places a premium on interrogating algorithmic outputs, not merely operating them. Second, AI-related expectations are rarely labeled as such; instead, firms express them through terms like "data analysis," "digital audit tools," and "automation," which appear in 60% of postings, compared with only 33% that mention AI by name. This terminological gap has implications for how educators frame curriculum content. Third, while all five universities reviewed offer courses such as IT Audit, Big Data Analysis and Audit, and Intelligent Accounting, the course titles alone do not reveal whether these offerings are core requirements, whether they involve hands-on tool training, or whether they integrate critical thinking in digital contexts. The study therefore identifies a critical need for deeper curriculum analysis and closer academia-industry collaboration.

5. Visible but Silent

Liao Yajun

Learning Sciences and Assessment, National Institute of Education

Abstract:

Key Literature & Research Gaps

Current discourse lauds AI's potential for personalized learning and adaptive systems, yet is dominated by techno-solutions. While industry pushes for scalable AI tools and academia focuses on pedagogical efficacy, the literature neglects the silent interface (the invisible data trails), algorithmic biases, and the passive consumption of AI-generated content. A critical gap exists in understanding how this silence (lack of student agency, opaque decision-making) impacts critical thinking and equity, despite the "visible" proliferation of AI dashboards and metrics.

Research Questions

RQ1: How does the "silent" algorithmic curation of learning content shape students' metacognitive awareness and autonomy?

RQ2: What are the friction points between industry-driven AI efficiency metrics and academic goals of deep, constructivist learning?

RQ3: How can we design a "visible" pedagogical framework that translates opaque AI processes into actionable dialogue for educators and learners?

Research Plan / Key Findings (Hypothesized)

We propose a mixed-methods Design-Based Research (DBR) study over 18 months, involving three university-industry partnerships.

Phase 1 (Audit): Analyze learning analytics dashboards to map "silent" variables (click-streams, time-on-task) against summative performance.

Phase 2 (Intervention): Co-design a "translucency" module that visualizes AI decision-logic to students in real-time.

Phase 3 (Validation): Implement a quasi-experimental trial (N=400) measuring critical thinking gains (pre/post) and qualitative interviews on student agency.

Key Finding (Projected): Preliminary data suggests that making AI's "silent" weighting explicit significantly reduces performance anxiety but increases cognitive load, necessitating a balanced triadic governance model among academia, tech, and industry.

6. From River Observations to AI: Understanding Sediment Yield in Tropical Monsoon Rivers

Huang Jiachun

Humanities and Social Studies Education, National Institute of Education

Abstract: River sediment strongly affects floodplains, reservoirs, ecosystems, and coasts, but sediment transport remains poorly understood in many tropical monsoon rivers. This study establishes a regional baseline of water and suspended-sediment export across 12 river basins in Peninsular Malaysia using long-term national monitoring records. Sediment yield varied strongly among rivers, ranging from about 50 to 587 t km⁻² yr⁻¹ at basin outlets. Sediment transport was also highly concentrated in time: the three-month peak-flow period transported 38–81% of annual suspended sediment. Water yield explained only part of the differences in sediment yield among basins. Catchment slope and built-up area were more important controls, showing that river sediment is influenced by both natural landscape conditions and human disturbance. These results provide an observational baseline for understanding tropical river systems. The next step is to examine whether artificial intelligence and machine-learning models can learn nonlinear relationships among rainfall, river flow, topography, land use, and sediment transport. Such models could complement process-based approaches and help identify when and where unusually high sediment transport is likely to occur.

7. A data-informed framework for individualized exercise prescription in post-stroke gait rehabilitation- A pilot randomized feasible study

Li Jiaqi

The Hong Kong Polytechnic University

Abstract: Most individuals with stroke experience walking difficulties, and standard exercise training may not address heterogeneous gait deviations.

Methods

Individuals with post-stroke lower-extremity impairments were randomized to a data-informed intervention or control group. In the experimental group, exercises were prescribed using a predefined biomechanical mapping framework based on lower-limb gait deviations identified through inertial measurement units and Statistical Parametric Mapping. The control group received standardized task-oriented training. Both groups completed five exercises per session across 12 supervised sessions over 6 weeks. Feasibility and acceptance were evaluated based on attendance, safety, and participant feedback. Gait, clinical, and sEMG outcomes were analyzed exploratorily using aligned rank transform models, supplemented by selected sensitivity and participant-level analyses.

Results

Seventeen individuals registered, sixteen met the eligibility criteria, and all enrolled participants completed the intervention and post-intervention assessment. All participants attended every scheduled session, with no dropouts or intervention-related adverse events. Participant feedback indicated good acceptance of the intervention. For the exploratory outcomes, the experimental group showed a greater numerical reduction in Gait Profile Score, although the group-by-time interaction was not significant ($p = 0.187$). Exploratory group-by-time interactions favoring the experimental group were observed for the composite gait deviation score ($p = 0.014$), FMA-LE ($p = 0.032$), TUG ($p = 0.008$), and TIS ($p = 0.006$). No significant interactions were observed for SIPSO, SPPB, BBS, rectus femoris S-iEMG ($p = 0.294$), or tibialis anterior S-iEMG ($p = 0.596$).

Conclusions

This pilot feasibility study demonstrated that the data-informed assessment-to-prescription workflow was feasible, safe, and acceptable under supervised laboratory conditions. This was supported by complete attendance, no dropouts, full intervention completion, and positive participant feedback. The exploratory gait, clinical, and sEMG findings warrant further investigation in a larger, adequately powered trial.

8. Sustainable Food Security Through the Implementation of Integrated Aquaponics Systems

Sari Rahayu Wijayanti

Universitas Negeri Surabaya/ State University of Surabaya

Abstract: Global food security is currently facing challenges due to agricultural land conversion, limited water resources, and the impacts of climate change. Aquaponics—an innovative technology that integrates aquaculture (aquatic organisms) and hydroponics (soilless plant cultivation) within a closed-loop ecosystem—emerges as a primary solution for efficient and environmentally friendly food production. This study aims to analyze the potential, efficiency, and operational challenges of implementing aquaponics systems to support sustainable food security. Utilizing a literature review and descriptive-analytical approach, this research evaluates water recirculation mechanisms and the role of nitrifying bacteria in converting fish waste into organic nutrients for plants. The findings indicate that aquaponics can reduce water usage by up to 90% compared to conventional agriculture, optimize limited land utilization (such as in urban farming), and produce two distinct food commodities—animal protein and vegetables—simultaneously without synthetic chemical fertilizers. However, constraints such as high initial investment costs, stable electrical power requirements, and dependency on commercial fish feed remain key operational challenges. This study concludes that aquaponics holds strategic prospects for strengthening community-level food security independently and sustainably, and recommends further integration with renewable energy technologies.

9. Diversity Metrics and Inclusive Cultures in AI-Enabled Tech Workplaces: Implications for Education and Gender Innovation

Edmongino J. Camacho

Philippine Women's University

Abstract: Artificial intelligence (AI) is transforming technology workplaces, creating new demands for workforce competencies, inclusive organizational practices, and responsible innovation. This study examined how workforce diversity and Diversity and Inclusion (D&I) program implementation influence inclusive organizational culture in Philippine technology companies, with implications for education, workforce preparation, and gender-responsive innovation in AI-enabled environments. Using a descriptive-correlational design, data were collected from 378 respondents across ten strata of medium- to large-scale technology firms. The study assessed workforce diversity, D&I implementation, and inclusive culture using measures of psychological safety, equity, openness, and leadership support. Results revealed high workforce diversity ($M = 3.66$, $SD = 0.41$), strong D&I implementation ($M = 3.70$, $SD = 0.41$), and highly inclusive culture ($M = 3.82$, $SD = 0.46$). Workforce diversity and D&I implementation were strongly correlated with inclusive culture ($r = .80$ and $r = .90$, $p < .001$). Global Market Orientation partially mediated these relationships, while Leadership Inclusivity significantly moderated the diversity–culture relationship ($B = 0.19$, $p < .001$). The resulting Gender-Based Innovation in Neo-Liberal Organization (GINO) Model provides an academia–industry framework for integrating diversity, inclusive leadership, gender-responsive innovation, and future-ready workforce development, supporting equitable and human-centered transformation in AI-enabled technology industries.

10. Leveraging Ensemble Machine Learning to Predict Resilience among Aging Entrepreneurs in the Global Economy

Fernando I. Talion

Conrado Benitez Institute for Business Education, Philippine Women's University

Abstract: Population aging is reshaping global entrepreneurship, creating a need for education, technology, and industry to support older adults in sustaining entrepreneurial activity. While artificial intelligence (AI) and machine learning provide powerful capabilities for identifying patterns associated with entrepreneurial resilience, their application to aging entrepreneurship remains underexplored. This study applies ensemble machine learning to predict resilience factors among aging entrepreneurs and examines how AI-driven insights can inform entrepreneurship education, policy, and industry support. Using data from the Global Entrepreneurship Monitor (GEM) Adult Population Survey (APS), boosted trees and random forests are employed to predict entrepreneurial resilience based on psychological, socioeconomic, demographic, and institutional factors. Feature engineering and rigorous model validation are conducted, with predictive performance benchmarked against traditional logistic regression. Findings identify critical factors associated with sustained entrepreneurial activity and demonstrate the superior predictive capability of ensemble models. The study further highlights how AI-generated evidence can support lifelong entrepreneurship education through targeted digital upskilling, mentoring, competency development, and responsive learning interventions for aging entrepreneurs. By bridging academic research, machine learning technology, and industry practice, the study demonstrates the potential of responsible AI to strengthen inclusive, evidence-based educational strategies and entrepreneurial ecosystems that promote resilience, innovation, and sustainable economic participation among aging populations worldwide.

11. Operational Performance Evaluation Of River Basin Flood Forecasting Warning Center in the Province of Pampanga

Rizza D. Gerente

Our Lady of Fatima University (OLFU), San Fernando, Pampanga

Abstract: Climate change, rapid urbanization, and increasingly extreme weather events underscore the need for reliable flood forecasting and early warning systems to protect vulnerable communities. This study evaluates the operational performance of the River Basin Flood Forecasting and Warning Center in Pampanga Province, Philippines, focusing on its capacity to generate timely, accurate, and actionable flood information. Using a descriptive-evaluative research design, the study assesses key dimensions of operational performance, including monitoring and forecasting capability, warning dissemination, technological infrastructure, human resource competency, inter-agency coordination, and community responsiveness. Particular attention is given to the integration of artificial intelligence, automated hydrometeorological monitoring, data analytics, and decision-support technologies in strengthening flood forecasting and risk communication. Data are gathered from technical personnel, institutional stakeholders, and relevant operational records to identify performance strengths, technological gaps, and capacity-building requirements. The study further examines how partnerships among government, academia, technology providers, and communities can strengthen professional training and promote evidence-based disaster preparedness. Findings are expected to inform a proposed AI-enabled operational improvement framework that integrates technological innovation, workforce development, and institutional collaboration. The framework aims to enhance forecasting reliability, warning effectiveness, organizational resilience, and sustainable disaster risk reduction in flood-prone communities.

3MT ® Competition

Time: 10:00 to 11:30

Venue: LT2

1. Code, Care & Childhood – Can AI Reset Education?

Wang Ziwei

Policy, Curriculum, and Leadership, National Institute of Education

Abstract: Driven by rapid AI transformation, Shenzhen’s education sector embraces intelligent tools to boost learning efficiency, yet faces the risk of diminishing students’ independent thinking and emotional engagement. Grounded on frontline teaching practice and the innovative model of Shenzhen Mingde Experimental School, this 3MT presentation redefines AI-driven educational reset through three core dimensions: Code, Care, and Childhood, integrating two authoritative frameworks—the 21st-century 4C competencies and Professor Peng Kaiping’s ACE positive education framework (Aesthetics, Creativity, Empathy).

Code represents AI hardware and digital systems. Mingde’s AI sports museum, robotics facilities and STREAM laboratories effectively foster students’ digital literacy and 4C core abilities. However, over-reliance on AI tends to cultivate passive answer-getters, restricting in-depth critical and creative thinking.

Care aligns with the ACE framework’s core *Empathy*, referring to interpersonal connection and loving energy exchange. Mingde’s inclusive, diverse educational ecosystem inspires students to solve real social issues. Iconic practices, including the elderly-friendly “Smart Grocery Basket” and the government-adopted Kunpeng Path wildlife corridor, prove that empathy-driven care is the fundamental engine of authentic innovation and critical thinking.

Childhood anchors the ultimate goal of educational reset. By providing free exploration and child-centred STREAM courses, the school preserves children’s innate curiosity and creativity, evidenced by students’ independent invention patents.

This study concludes that AI alone cannot reset education. Code provides tools, Care nourishes human motivation, and Childhood sets our bottom line. Only integrating technological Code, humanistic Care and child-centred Childhood can balance 4C and ACE cultivation, achieving genuine, humane educational transformation for future learners.

2. Leveraging Artificial Intelligence and Gamification to Enhance Exercise Physiology Education: The Gametize Platform

Ng Yew Cheo

Physical Education and Sport Science, National Institute of Education

Abstract: The integration of artificial intelligence (AI) and gamification into higher education presents new opportunities to enhance teaching and learning in exercise physiology.

Purpose

To examine the use of AI-enabled platform as an innovative AI-tool designed to support student engagement, understanding, and retention of complex physiological concepts and laboratory procedures.

Methods

AI-generated avatar videos of the professor were developed to deliver concise instructional content on key practical components in this pilot educational innovation. Additional videos also included maximal oxygen uptake ($\text{VO}_{2\text{max}}$) treadmill testing, blood lactate sampling and analysis using a YSI analyzer, the Wingate anaerobic test, and body composition assessment via bioelectrical impedance analysis (BIA). Each module was accompanied by quizzes, where a class of 25 postgraduate students had to complete them and were awarded points for correct responses, reinforcing learning through gamified feedback mechanisms. The platform's accessibility across mobile and web-based devices allowed students to engage in self-paced and repeated learning beyond scheduled class time. Grounded in principles of cognitive load theory, microlearning, and retrieval-based practice, Gametize facilitated active engagement and improved knowledge consolidation.

Results

Findings from the pilot study indicated positive student perceptions, including increased motivation, enhanced understanding of laboratory procedures, and appreciation for the brevity and clarity of instructional videos. Students also valued the ability to revisit content and receive immediate feedback on their performance. Conclusion

Overall, the integration of AI and gamification through the AI-enabled platform demonstrated significant potential to improve educational practice in exercise physiology by promoting flexible, student-centered, and data-driven learning environments. This approach offers a scalable model for enhancing both theoretical comprehension and practical competency in higher education.

3. An Action Research on the Application of Generative AI Tools in an Industry–Academia Collaboration Program

Wen-Hsiu Liu

Asian Languages and Cultures, National Institute of Education

Abstract: With the implementation of Taiwan's New Southbound Policy, an increasing number of students from Southeast Asian countries have chosen to pursue higher education in Taiwan through industry–academia collaboration programs that combine academic learning with workplace training. However, differences in language proficiency, cultural background, and learning needs make it challenging for instructors to design instructional materials that effectively support international students. The rapid development of Generative Artificial Intelligence (Generative AI) provides new opportunities for creating more adaptive, flexible, and learner-centered teaching materials that better address these diverse educational needs.

This study adopts an action research approach to investigate the application of Generative AI tools in instructional design within an industry–academia collaboration program jointly conducted by a Taiwanese university and FamilyMart. Generative AI tools will be used to support the development and continuous refinement of instructional materials throughout the teaching process. Data will be collected through classroom observations, students' learning feedback, learning artifacts, and the instructor's reflective journals. By following iterative cycles of planning, implementation, observation, and reflection, the study aims to explore how Generative AI can assist instructors in designing instructional materials that better meet the learning needs of Southeast Asian students.

The study is expected to propose practical instructional design strategies for integrating Generative AI tools into industry–academia collaboration programs. The findings are expected to provide practical implications for supporting international students, enhancing cross-cultural teaching practices, and promoting the effective integration of Generative AI into higher education.

4. It's so hard to explain the use of AI for education research

Lye Yu Min

Psychology and Child and Human Development, National Institute of Education

Abstract: In social and education sciences, empirical research traditionally involves the framing of a theoretical model, hypothesising variable relationships, collecting data, and drawing inferences about (i.e. explaining) how specific predictors influence an outcome of interest. On the other hand, the field of data and computer science, which gave birth to artificial intelligence (AI), focuses on predicting future outcomes. But the word "predict" is used differently in these fields. Mention "AI" today, and most people immediately picture large language models, conversational chatbots, humanoid robots, or intelligent autonomous agents. In reality, underlying techniques like machine learning and deep learning are essentially statistical models that can be surprisingly explainable.

My research aims to cross these two paradigms. By working with rich datasets and employing both data-driven and theory-driven approaches, I demonstrate how data science techniques can be practically applied within social and education science contexts without sacrificing theoretical depth.

Bridging these disciplines, however, is much easier said than done. It has been a challenging 2.5-year journey explaining this integration to my academic discourse community, e.g. reassuring social and education researchers that machine learning is not a black box, while convincing data scientists that complex human behaviours require solid theoretical foundations. Ultimately, uniting data science traditions with social science purpose may allow us to uncover richer, more actionable insights into how children develop.

5. Towards the Balance Between Specificity and Repetition

Goh Khee (Wu Qi)

Physical Education and Sport Science, National Institute of Education

Abstract: Sport training design today aims to move away from 'traditional' approaches that involve highly repetitive but often non-specific training tasks. Instead, coaches are encouraged use a representative learning design (RLD) to improve the specificity of training tasks, hence allowing for more accurate simulations of game context and subsequently, better player performance. At present, the application of RLD is often seen as small-sided and/or simulated games, prior research into such training tasks have found them to be just as effective if not, at times, more effective than training tasks developed with 'traditional' approaches. However, despite seeing success, it was also found that such tasks alone may not be sufficient for learning, i.e., simply playing the game is insufficient for player development due to the lack of repetitions for enough exposure to the performance contexts that training tasks intend to simulate.

Such an issue finds that methods of task design in coaching seem to forgo either the specificity or repetitions of the task. Therefore, the presented study is conducted to find out how the rule induction framework can be used to derive sets of correlated constraints from competitive matches to (i) compare against sets of correlated constraints derived from training, and (ii) identify what exactly makes up game contexts for eventual application to the development of training tasks that can be repeated while maintaining its specificity to game contexts.

Results have found that not only are experienced constraints in training vastly different from that of competitive matches, but they also often have elements that are almost simulating game context. Additionally, precise points of differences in constraints between those in competitive matches and in training can be identified to eventually better inform how training tasks can be better modified and/or developed to be specific while allowing sufficient repetitions.

6. Speak, Self-Reflect, Succeed-AI Powered Oracy Feedback

Kiranjeet Kaur

Policy, Curriculum and Leadership, National Institute of Education

Abstract: To what extent, does Artificial Intelligence (AI) generated feedback help students improve on their oracy and the role of teachers in this AI powered teaching tool?

Educators face time constraint in testing oracy in every student in a classroom. Often, only a few volunteers will present to serve as exemplars or role models. On many occasions, teachers do feedback that testing oracy is time consuming and it's often not considered as a priority unless the scheme of work for that term is covered. In addition, teachers also face challenges in providing constructive feedback for each student to make refinements for improvement. Usually a general feedback is provided to the class. How then do we provide customised feedback for improvement? By generating AI powered feedback, students are able to receive detailed feedback in alignment with rubrics and grow to be self-directed to use feedback for refinement. Teachers act as facilitators to ensure feedback provided is accurate and according to students' capacity. This can be done based on teachers prior knowledge of students potential and the rubrics for assessment. Feedback provided can also include suggested ideas to discuss about in student's presentation which will also allow student to widen their perspectives and develop critical thinking and communication skills.

7. Making Sense of Language Value: Language Ideologies in Two Undergraduates' Narratives

Zhang Xiangwan

English Language and Literature, National Institute of Education

Abstract: Language ideologies concern the beliefs and assumptions through which people understand languages, their use, and their social meanings (Woolard & Schieffelin, 1994). Rather than being uniform or fixed, language ideologies may vary across social positions and contexts (Irvine, 2022). While language ideology research has illuminated the broader social and institutional processes through which meanings and values are attached to languages, there remains value in examining how individuals make sense of their language ideologies through lived experiences, particularly as they move across different educational and social contexts. This study asks: How do two undergraduates studying in Singapore make sense of the development of their language ideologies through their lived experiences? Adopting a narrative inquiry approach, the study draws on multiple semi-structured interviews with one Singaporean and one Malaysian undergraduate studying in Singapore.

The findings suggest that participants' language ideologies are closely associated with the practical value they attribute to different languages. Their narratives illustrate how understandings of language value develop through schooling, family life, social interaction, cross-border mobility, and changing educational environments. In particular, changes in educational and social contexts can reshape the opportunities and needs for using particular languages and, consequently, the meanings and values participants attach to them. These findings highlight the importance of attending to students' lived experiences in language education and offer insights for language policy. They suggest that efforts to promote heritage languages should extend beyond formal instruction and consider whether learners have meaningful opportunities and social contexts in which these languages can be actively used.

8. When AI Isn't Enough: Why Structured Scaffolding Matters

Xu Yuchen

Learning Sciences and Assessment, National Institute of Education

Abstract: Generative artificial intelligence (GAI) is increasingly being integrated into programming education, yet its educational benefits are not automatic. While existing research has largely focused on GAI as a tool for generating code or providing answers, less attention has been given to how instructional design shapes its impact on students' deeper learning. In particular, it remains unclear whether teacher-designed scaffolding can enhance the effectiveness of GAI-supported reflective learning.

This mixed-methods quasi-experimental study investigates whether and how structured GAI scaffolding supports students' metacognition and computational thinking in programming education. Three instructional conditions were compared among 120 Grade 10 students: Traditional Reflective Teaching, GAI-Assisted Reflective Teaching with unrestricted AI use, and Structured GAI-Assisted Reflective Teaching using a teacher-designed six-step reflection protocol, staged prompts, and a rubric for self-assessment. The study addresses three questions: whether GAI-assisted reflection improves learning outcomes, whether structured scaffolding amplifies the effects of GAI, and whether reflection quality explains these effects.

The findings show that structured GAI-supported reflection significantly outperformed both traditional reflection and unstructured GAI use in metacognition and computational thinking. Reflection quality also fully mediated these effects. These findings suggest that AI alone is not enough: its educational value depends on how teachers structure students' interaction with it. The study offers practical principles for designing effective AI-integrated programming instruction.

9. Reading the Past to Prepare for Future Floods

Huang Jiachun

Humanities and Social Studies Education, National Institute of Education

Abstract: How can we prepare for the next extreme flood if our observations cover only a small part of the past? Peninsular Malaysia experiences recurrent floods driven by intense tropical rainfall, monsoons, landscape conditions, and human activities. Yet most river monitoring records cover less than a century. This makes it difficult to understand rare, extreme floods and how they may change in the future. My PhD addresses this problem by connecting the present, past, and future. First, I use long-term river observations across Peninsular Malaysia to understand how water, sediment, and floods vary between river basins and what controls these differences. Second, I go beyond instrumental records by studying sediment cores and other geological evidence preserved in floodplains. These natural archives can reveal floods that occurred hundreds or thousands of years ago. Finally, I will combine evidence from past floods with modern observations, climate projections, and hydrological modelling to investigate how future flood frequency and magnitude may change. By connecting what rivers are doing today, what they have done in the past, and what they may do in the future, this research aims to provide a longer and more reliable view of extreme flooding and support better flood preparedness in tropical regions.

10. Connecting Education, Research, and Innovation: How Hong Kong Universities Help Build an International Talent Hub in the Age of AI

Hong Chang

Policy, Curriculum, and Leadership, National Institute of Education

Abstract: The rapid development of artificial intelligence (AI) is reshaping knowledge production, talent development, and technological application, increasing the importance of higher education in connecting academia, technology, and industry. Existing studies have often examined the effects of AI on education, innovation, and industrial application separately, leaving insufficiently understood how AI reconfigures the relationships among education, research, and innovation. Drawing on the Knowledge Triangle framework, this qualitative case study analyses policy texts, official plans, statistical data, and related materials to examine how AI affects the education-research, research-innovation, and education-innovation linkages through which Hong Kong universities contribute to building an international talent hub.

The findings show that these linkages form a synergistic mechanism connecting talent development, knowledge production, and innovative application. AI provides both the technological context for this mechanism and new means through which it operates, thereby strengthening organisational coordination and continuous feedback among education, research, and innovation. The pressures and opportunities generated by AI are translated into concrete action through the guidance of China's national strategies, policy translation by the Hong Kong Special Administrative Region Government, and the organisational responses of universities. This process enables universities to cultivate and attract talent through education, bring talent together through research, and develop and retain talent through innovation. The study thus extends the Knowledge Triangle from a relatively static structure into a dynamic model of continuous interaction under conditions of rapid technological change.

11. Seeing the Game Beyond the Ball: Using Artificial Intelligence to Decode Netball Performance

Amandeep Kaur Chahal

Physical Education and Sport Science, National Institute of Education

Abstract: Traditional netball statistics tell us what happened in a match—such as goals scored, turnovers, or interceptions—but rarely explain how those moments were created. Coaches often rely on subjective observation to understand team movement and tactical organisation because tracking every player's movement throughout an entire game is both difficult and time-consuming. My research aims to develop the first automated player-tracking system specifically designed for netball using computer vision technology. Using a single match video, the system will detect every player, identify their team and playing position, and map their movements onto a two-dimensional representation of the court. Once validated against manual tracking, these positional data can be used to quantify collective tactical behaviours such as team spacing, movement patterns, and defensive organisation. By moving beyond isolated match statistics, this research aims to provide coaches with objective insights into how teams create scoring opportunities, generate defensive gains, and ultimately make more informed, evidence-based tactical decisions in elite netball.

12. A Chest-Mounted Edge AI Assistant for Peripersonal Navigation and Real-Time Caregiver Telemetry

Chad Vincent Florentino

Research and Development Office, Philippine Women's University

Abstract: Assistive navigation technologies have emerged as promising solutions for improving mobility and environmental awareness among visually impaired individuals. This study evaluates the operational performance of an existing chest-mounted edge artificial intelligence (AI) assistive navigation system that integrates real-time object detection, spatial localization, audio feedback, and cloud-based caregiver telemetry. Using an Evaluation Research approach, the system was assessed through structured alpha testing under controlled indoor and outdoor environments. Performance was measured using confusion matrix analysis and standard object detection metrics, including Precision, Recall, Accuracy, and F1-score. Experimental results demonstrated a Precision of 98.28%, Accuracy of 96.95%, Recall of 48.72%, and F1-score of 65.14%. The findings indicate that the system provides highly reliable object recognition with a low false-positive rate while maintaining effective environmental awareness capabilities. Although recall performance suggests opportunities for further optimization, the results validate the operational feasibility of integrating edge AI and cloud-based telemetry within a wearable assistive navigation framework. The study contributes empirical evidence regarding the effectiveness of intelligent assistive technologies for supporting independent mobility and caregiver situational awareness.

13. To Inquire, Not Merely to Answer — Rethinking Knowledge Discovery Under the AI Tide

Li Siyao

Nanjing Normal University, China

Abstract: For a long time, teachers have been regarded as the authoritative transmitters of disciplinary knowledge, and classroom instruction in China has consistently focused on how to deliver established content efficiently and accurately. However, the rise of generative AI is disrupting this unilateral logic. When knowledge can be generated instantly, the core of teaching and learning is no longer the replication of “correct answers,” but rather the knowledge discovery process that teachers and students undertake together when facing real-world, complex, and ill-defined problems. This shift poses profound challenges to the professional role of teachers.

Drawing on a provincial training program for secondary school geography teachers in field teaching competence, we observed and analyzed teachers’ cognitive decision-making patterns in authentic contexts. Our findings reveal that teachers exhibit a strong preference for “accurate judgment”—they tend to seek a single correct geographical explanation, definitive classification labels, and deterministic causal chains, while rarely attempting to generate multiple, open, or even tentative inferences based on limited evidence. This cognitive inertia creates tension with the uncertain thinking demanded by the AI era, and exposes a new blind spot in teacher professional development.

We argue that teacher competence in the AI era should go beyond mastering tools; it must shift toward the ability to harness plausible reasoning—that is, tolerating ambiguity, weighing evidence, generating hypotheses, and guiding students to participate in this process. This study suggests that teacher training needs to move from “pursuing accuracy” to “training reasoning,” so as to redefine teachers’ roles in knowledge discovery and provide a new cognitive framework for teacher-student collaboration in future education.

14. Prompt Engineering Model for Multicultural Psychological First Aid: Training *musyrif* and *musyrifah* capacity as a Student Caregivers in Indonesian *pesantren*

Hengki Tri Hidayatullah, Jasmine Nurul Izza (*presenter*)

Universitas Negeri Malang, Indonesia

Abstract: Life in Islamic boarding school (*pesantren*) requires students (*santri*) to adapt to a complex social, academic, and spiritual environment. Such conditions often give rise to psychological challenges such as stress, loneliness, and homesickness. Which are typically first addressed by *musyrif* and *musyrifah*, the daily caregivers. However, most of them lack of formal training in counselling communication skills or Psychological First Aid (PFA). This study aims to develop and test KOMUNIKA v1.1 Multicultural SAFE+, a prompt-engineering and Large Language Model (LLM)-based simulation system designed to train multicultural counselling skills for *musyrif* and *musyrifah*. The study employed a qualitative approach using the design-based research (DBR) model during the initial two stages: (1) contextual problem and training needs analysis, and (2) prototype design and preliminary testing based on empirical data. Data were collected through in-depth interviews, form-based survey, and focus group design (FGDs) involving eleven *musyrif/musyrifah* and two *pesantren* principals in East Java. Data were analysis using Brain and Clarke's thematic analysis, and performance evaluation was conducted through rubric-based simulations within the KOMUNIKA system. The first phase revealed six main themes of students' psychological challenges (emotional adaptation difficulties, academic stress, interregional conflicts, identity confusion, religious guild, and social withdrawal). In the second phase, the KOM-02 simulation case (Homesickness and Loneliness in Dormitories) was tested by three *musyrif/musyrifah* without counselling backgrounds. Results indicate average skills scores ranging between 2.17-1.96 (Fair-good category), with the highest dimensions being Cultural Bridging (M=2.94) and Acceptance & Non-Judgement (M=3.04), with the lowest were Open Questioning (M=1.72) and Clarification (M=2.07). The finding demonstrated that KOMUNIKA V1.1 is an effective AI-based training medium capable of facilitating reflective learning and culturally responsive feedback for *musyrif* and *musyrifah*. This model holds potential as an innovative counselling education tool in *pesantren*, strengthening caregivers' readiness to deliver safe, empathetic, and culturally sensitive psychological support.

15. Mesoscale Convective Systems Tracking Database over SEA

Zhou Xuyang

Humanities and Social Studies Education, National Institute of Education

Abstract: Thunderstorms, tropical storms and squall lines are well-known heavy-precipitating systems, which are also characteristic types of mesoscale convective systems (MCSs). They tend to produce heavy rainfall in tropical Southeast Asia (SEA). Feng et al. (2021) demonstrated that the combination of infrared brightness temperature (Tb) and precipitation features (PFs) enables the object-based detection and tracking of MCSs, while Crook et al. (2024) showed that MCSs contribute substantially to regional rainfall. However, the relationships between MCS lifecycle and precipitation characteristics across the complex land–sea environment of SEA remain insufficiently quantified. Also, due to the long lifecycle of some tropical systems, it is necessary to enhance the tracking of spatial changes.

An hourly, 10-km MCS tracking dataset is being developed over SEA using PyFLEXTRKR, merged geostationary infrared brightness temperature combined with GPM IMERG precipitation data. Cold-cloud systems are tracked through time and linked to embedded precipitation features. A hierarchical methodology has been implemented to assist convective cell tracking, especially for fast-moving objects with strong background winds. MCSs associated with tropical cyclones are separated using IBTrACS storm positions and radius-of-outer-closed-isobar information. Preliminary processing has identified all the MCS tracks during 2020-2024. Subsequent analysis will quantify MCS and non-MCS contributions to percentile-based precipitation extremes and relate extreme rainfall to MCS life-cycle stage, intensity, duration, and propagation. The resulting regional climatology will provide an observational basis for improving extreme-rainfall risk assessment and model evaluation in SEA.

CrossSolve Showcase

Time: 15:00 to 16:00

Venue: LT2

Team 1: Koh Wei Xun; Ni Youhua; Varsha Vignesh; Han Chouyu

Preparing Our Students for the Age of AI

Team 2: Erti Hamimi; Liu Xiaoyin; Jia Yichen; Song Mingyang

The Flight Simulator for Teachers: An AI-Powered Virtual Classroom for Teaching Practice, Adaptation, and Improvement

Team 3: Rahmati Putri Yaniafari; Feng Tongyang; Li Jiangya; Alistair Peacock

Beyond the Right Answer: Closing the Transfer-of-Learning Gap Through Applied, Curriculum-Linked Practice

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